FITRIA ROZI

Junior Software Engineer | AI & Computer Vision Enthusiast

fitriarozi03@gmail.com | 082384117474 | Batam, Kepulauan Riau
LinkedIn | GitHub | Live Demo: OnionBot AI Platform

PROFESSIONAL SUMMARY

Fresh graduate in Physics Instrumentation with a strong foundation in programming, data structures, algorithms, and object-oriented programming, built through self-driven AI projects beyond coursework. Experienced in building end-to-end software systems: from deep learning models and backend APIs (FastAPI) to production deployment (Docker, Hugging Face Spaces). A fast, self-motivated learner who enjoys solving complex technical problems systematically and is eager to explore emerging technologies such as AI applications and LLM-based systems.

PROJECTS

OnionBot AI Platform — Onion Disease Detection & Smart Agriculture Chatbot huggingface.co/spaces/jifua19/onion-disease-ai

Personal Project | Python, FastAPI, ONNX Runtime, OpenCV, Docker, Mini-RAG Chatbot

- Built a full-stack AI system for onion leaf disease detection from uploaded images, with confidence scoring and treatment recommendations based on an agricultural knowledge base
- Developed a FastAPI backend serving a deep learning model deployed in ONNX format for lightweight, fast inference
- Designed a mini-RAG (retrieval-augmented) chatbot combining a local knowledge base with Wikipedia search to answer agriculture-related questions contextually
- Containerized the application with Docker and deployed it to Hugging Face Spaces, including cross-platform (Windows-to-Linux) debugging for case-sensitivity, dependency, and Git LFS issues
- Built supporting features: prediction history (SQLite), analytics dashboard, and user authentication system

Tomato Leaf Disease Classification

Research Intern Project | TensorFlow, MobileNet, ONNX, Hailo AI, Raspberry Pi

- Developed a CNN model using MobileNet for tomato leaf disease classification, achieving 98% accuracy on a 10,000-image dataset
- Converted the model from ONNX to Hailo HEF format for edge device deployment
- Deployed the model on a Raspberry Pi 5 with real-time inference using a Raspberry Pi camera

Wave Simulation FFT — Python

- Simulated wave behavior using Fast Fourier Transform with 3D visualization (Matplotlib) and frequency-domain analysis

Electromagnetic Wave Simulation — FDTD

- Simulated 1D electromagnetic wave propagation using the FDTD method, including reflection and transmission analysis

EXPERIENCE

Research Intern – MBKM BRIN *August 2024 – January 2025*

Photonics Research Center, Serpong

- Developed a deep learning model for plant classification using a TensorFlow-based CNN
- Processed a 10,000-image dataset from Kaggle and Mendeley, covering preprocessing, training, validation, and testing
- Evaluated model performance using confusion matrix, precision, recall, and ROC-AUC, achieving up to 98% accuracy
- Created training accuracy and loss visualizations for overfitting analysis

Basic Physics Laboratory Assistant *August 2023 – August 2025*

Basic Physics Laboratory, Universitas Andalas

- Guided students through physics experiments and data processing
- Processed experimental data using Excel and basic statistical analysis
- Prepared technical documentation and experiment procedures

EDUCATION

B.Sc. Physics Instrumentation *August 2021 – August 2025*

Universitas Andalas, Padang, West Sumatra | GPA: 3.57 / 4.00

SKILLS

Programming: Python, SQL, JavaScript (basic)

Computer Science Fundamentals: Data Structures, Algorithms, Object-Oriented Programming (OOP)

Machine Learning: TensorFlow, PyTorch, Scikit-learn, CNN, Transfer Learning

Computer Vision: OpenCV, Image Classification, Deep Learning
Backend & Web: FastAPI, REST API, Docker, SQLite, HTML/CSS/JS
Deployment: Raspberry Pi, ONNX, Hailo AI, Docker, Hugging Face Spaces, Linux
Data Processing: Pandas, NumPy, Matplotlib
Tools: Git/GitHub, Jupyter Notebook, Google Colab, VS Code

CERTIFICATIONS

- Python for Data Science, AI & Development — IBM — 2026
- Data Analysis and Visualization with Power BI — Microsoft — 2026
- Data Manipulation with Pandas — DQLab — 2026
- Exploratory Data Analysis with Python — DQLab — 2026
- Python Fundamental for Data Science — DQLab — 2026
- Data Visualization with Python Matplotlib — DQLab — 2026